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 Studierichting: Informatica
 Oefeningenreeks 4A nr. 39.5

$$\int \frac{3x-1}{(x^2+4x+5)^3} dx = \int \frac{3x-1}{((x+2)^2+1)^3} dx$$

$$u = x + 2 \Rightarrow x = u - 2 \text{ en } dx = du$$

$$\int \frac{3(u-2)-1}{(u^2+1)^3} du = \int \frac{3u-7}{(u^2+1)^3} du = 3 \int \frac{u}{(u^2+1)^3} du - 7 \int \frac{1}{(u^2+1)^3} du$$

Eerste integraal:

$$3 \int \frac{u}{(u^2+1)^3} du$$

$$t = u^2 + 1 \Rightarrow dt = 2u du \Rightarrow u du = \frac{1}{2} dt$$

$$\frac{3}{2} \int t^{-3} dt = \frac{3}{2} \cdot \frac{t^{-2}}{-2} = -\frac{3}{4(u^2+1)^2}$$

Tweede integraal:

$$7 \int \frac{1}{(u^2+1)^3} du$$

$$\text{Recurisie: } I_n = \frac{u}{2(n-1)(u^2+1)^{n-1}} + \frac{2n-3}{2n-2} I_{n-1}:$$

$$I_1 = \text{Bgsin}(u)$$

$$I_3 = \frac{u}{4(u^2+1)^2} + \frac{3}{4} I_2 = \frac{u}{4(u^2+1)^2} + \frac{3}{4} \left(\frac{u}{2(u^2+1)} + \frac{1}{2} \text{Bgsin}(u) \right)$$

Dus:

$$\int \frac{3x-1}{(x^2+4x+5)^3} dx = -\frac{3}{4(x^2+4x+5)^2} - 7 \left[\frac{x+2}{4(x^2+4x+5)^2} + \frac{3(x+2)}{8(x^2+4x+5)} + \frac{3}{8} \text{Bgsin}(x+2) \right] + c$$

References